Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Reviewing drafter formatting and how to save inputs in a callable location will help me complete the project. Concepts of loops, lists, and nested data are something I expect to use. I hope to improve all of my general programming skills

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

All of it is going to be difficult; setting up the dataclasses should be the easiest part.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Setting up the drafter ui and states will take the most time. I should focus on the input boxes and input storage first.

D. What distractions might interfere with your work, and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Other class work will be the biggest distraction; as long as I manage time properly, it shouldn't be too big an issue. I will need the textbook and past assignments available to work on the assignment.

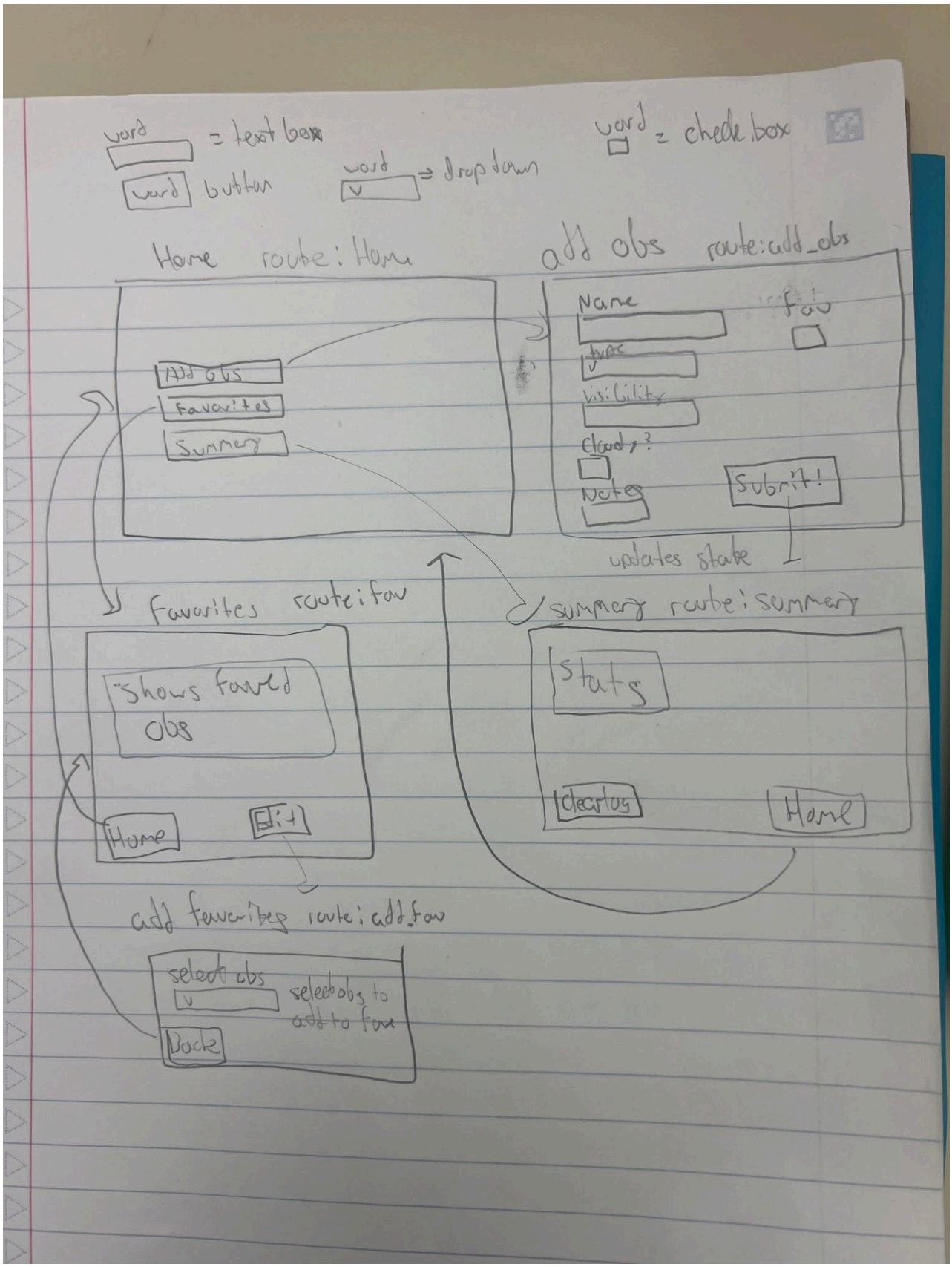
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


word = text box
word button
word ⇒ drop down
word = check box
Home route: Home
Add obs
Favorites
Summary
add obs route: add_obs
Name
fav
type
visibility
cloudy?
Notes
Submit!
updates state
Favorites route: fav
shows faved obs
Home
Edit
summary route: summary
stats
clear logs
Home
add favorites route: add_fav
select obs
select obs to add to fav
Back

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**To store observation data and return stats based on it, along with showing stored favorite observations**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Fields: Obs: list[Observation] Fav_obs: dict[str, Observation] Total_obs: int Sky_messasge: str Newest_obs: str
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Fields: Object_name: str Object_type: str Visibility: int Cloudy: bool Notes: str
Routes (minimum 7)	/ - homepage /add_obs - form for entering data

The names of your route functions	/submit_obs - processes obs and updates state /fav - shows fav objects /add_fav - adds an object to fav /summary - shows summaries of data /clear_obs - resets logs
Pages (minimum 5) A brief description of the pages in your site	Home page Add observations page Favorites page Summary page Add to favorites page
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Textbox: object name, type, notes Selectbox: visibility, edit favorite Checkbox: cloudy, favorite
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Visibility check: if visibility is bad, tell the user Cloudy check: if cloudy is True, update feedback text Object type update: update last_object_viewed
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	On /summary will iterate over the observations to calculate data

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

No ai

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.